Student: Xirong Zhang **Date:** 12/01/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 2.1

1. Compute each matrix sum or product if it is defined. If an expression is undefined, explain why. Let $A = \begin{bmatrix} 3 & 0 & -2 \\ 3 & -4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 6 & -4 & 1 \\ 2 & -4 & -2 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$, and $D = \begin{bmatrix} 3 & 5 \\ -2 & 3 \end{bmatrix}$.

$-2A$, $B-2A$, AC, CD

Compute the matrix product $-2A$. Select the correct choice below and, if necessary, fill in the answer box within your choice.

- **A.** $-2A = \begin{bmatrix} -6 & 0 & 4 \\ -6 & 8 & -4 \end{bmatrix}$ (Simplify your answer.)
- **B.** The expression $-2A$ is undefined because matrices cannot be multiplied by numbers.
- **C.** The expression $-2A$ is undefined because A is not a square matrix.
- **D.** The expression $-2A$ is undefined because matrices cannot have negative coefficients.

Compute the martrix sum $B-2A$. Select the correct choice below and, if necessary, fill in the answer box within your choice.

- **A.** $B-2A = \begin{bmatrix} 0 & -4 & 5 \\ -4 & 4 & -6 \end{bmatrix}$ (Simplify your answer.)
- **B.** The expression $B-2A$ is undefined because B and A have different sizes.
- **C.** The expression $B-2A$ is undefined because B and $-2A$ have different sizes.
- **D.** The expression $B-2A$ is undefined because A is not a square matrix.

Compute the matrix product AC. Select the correct choice below and, if necessary, fill in the answer box within your choice.

- **A.** $AC =$ (Simplify your answer.)
- **B.** The expression AC is undefined because the number of columns in A is not equal to the number of rows in C.
- **C.** The expression AC is undefined because the number of rows in A is not equal to the number of columns in C.
- **D.** The expression AC is undefined because the number of rows in A is not equal to the number of rows in C.

Compute the matrix product CD. Select the correct choice below and, if necessary, fill in the answer box within your choice.

- **A.** $CD = \begin{bmatrix} -1 & 11 \\ -8 & -7 \end{bmatrix}$ (Simplify your answer.)
- **B.** The expression CD is undefined because square matrices cannot be multiplied.
- **C.** The expression CD is undefined because matrices with negative entries cannot be multiplied.
- **D.** The expression CD is undefined because the corresponding entries in C and D are not equal.

2. Let $A = \begin{bmatrix} 2 & -1 \\ 5 & -4 \end{bmatrix}$. Compute $3I_2 - A$ and $(3I_2)A$.

$3I_2 - A = \begin{bmatrix} 1 & 1 \\ -5 & 7 \end{bmatrix}$

(Type an integer or decimal for each matrix element.)

$(3I_2)A = \begin{bmatrix} 6 & -3 \\ 15 & -12 \end{bmatrix}$

(Type an integer or decimal for each matrix element.)

3. Compute the product AB by the definition of the product of matrices, where $A\mathbf{b}_1$ and $A\mathbf{b}_2$ are computed separately, and by the row-column rule for computing AB.

$$A = \begin{bmatrix} -2 & 2 \\ 3 & 4 \\ 4 & -4 \end{bmatrix}, B = \begin{bmatrix} 3 & -2 \\ -3 & 4 \end{bmatrix}$$

Set up the product $A\mathbf{b}_1$, where $\mathbf{b}_1$ is the first column of B.

$$A\mathbf{b}_1 = \begin{bmatrix} -2 & 2 \\ 3 & 4 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

(Use one answer box for A and use the other answer box for $\mathbf{b}_1$.)

Calculate $A\mathbf{b}_1$, where $\mathbf{b}_1$ is the first column of B.

$$A\mathbf{b}_1 = \begin{bmatrix} -12 \\ -3 \\ 24 \end{bmatrix}$$

(Type an integer or decimal for each matrix element.)

Set up the product $A\mathbf{b}_2$, where $\mathbf{b}_2$ is the second column of B.

$$A\mathbf{b}_2 = \begin{bmatrix} -2 & 2 \\ 3 & 4 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

(Use one answer box for A and use the other answer box for $\mathbf{b}_2$.)

Calculate $A\mathbf{b}_2$, where $\mathbf{b}_2$ is the second column of B.

$$A\mathbf{b}_2 = \begin{bmatrix} 12 \\ 10 \\ -24 \end{bmatrix}$$

(Type an integer or decimal for each matrix element.)

Determine the numerical expression for the first entry in the first column of AB using the row-column rule. Choose the correct answer below.

- A. $-2(3) - 2(-3)$
- B. $((-2) + (3)) \cdot ((2) + (-3)$
- C. $((-2) - (3)) \cdot ((2) - (-3)$
- D. $-2(3) + 2(-3)$

Determine the product AB.

$$AB = \begin{bmatrix} -12 & 12 \\ -3 & 10 \\ 24 & -24 \end{bmatrix}$$

(Use integers or decimals for any numbers in the expression.)

4. If a matrix A is 4×2 and the product AB is 4×6, what is the size of B?

The size of B is 2 × 6.

5. Let $A = \begin{bmatrix} 3 & 4 \\ -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 6 \\ -3 & k \end{bmatrix}$. What value(s) of k, if any, will make AB = BA?

Select the correct choice below and, if necessary, fill in the answer box within your choice.

- **A.** k = -2 (Use a comma to separate answers as needed.)
- **B.** No value of k will make AB = BA

6. Let $A = \begin{bmatrix} 4 & -16 \\ -4 & 16 \end{bmatrix}$. Construct a 2×2 matrix B such that AB is the zero matrix. Use two different nonzero columns for B.

$B = \begin{bmatrix} 4 & 8 \\ 1 & 2 \end{bmatrix}$

7. Let A and B be arbitrary matrices for which the indicated product is defined. Determine whether the statement below is true or false. Justify the answer.

If A and B are 3×3 matrices and $B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix}$, then $AB = \begin{bmatrix} A\mathbf{b}_1 + A\mathbf{b}_2 + A\mathbf{b}_3 \end{bmatrix}$.

Choose the correct answer below.

- **A.** The statement is true. By the definition of matrix multiplication, if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then $AB = A\begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 + A\mathbf{b}_2 + \dots + A\mathbf{b}_p \end{bmatrix}$.
- **B.** The statement is true. By the definition of matrix multiplication, if A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the resulting matrix AB is the sum of the columns of A using the weights from the corresponding columns of B.
- **C.** The statement is false. The matrix $\begin{bmatrix} A\mathbf{b}_1 + A\mathbf{b}_2 + A\mathbf{b}_3 \end{bmatrix}$ is the correct size matrix, but the plus signs should be minus signs.
- **D.** The statement is false. The matrix $\begin{bmatrix} A\mathbf{b}_1 + A\mathbf{b}_2 + A\mathbf{b}_3 \end{bmatrix}$ is a 3×1 matrix, and AB must be a 3×3 matrix. The plus signs should be spaces between the 3 columns.

8. Let A and B be arbitrary matrices for which the indicated product is defined. Determine whether the statement below is true or false. Justify the answer.

The second row of AB is the second row of A multiplied on the right by B.

Choose the correct answer below.

- **A.** The statement is false. The second row of AB is the second row of A multiplied on the left by B.
- **B.** The statement is true. Every row and column of AB is the corresponding row and column of A multiplied on the right by B.
- **C.** The statement is true. Let $\text{row}_i(A)$ denote the ith row of matrix A. Then $\text{row}_i(AB) = \text{row}_i(A)B$. Letting $i = 2$ verifies this statement.
- **D.** The statement is false. Let $\text{column}_i(A)$ denote the ith column of matrix A. Then $\text{column}_i(AB) = \text{column}_i(A)B$. The same is not true for the rows of AB.

9. Let A, B, and C be arbitrary matrices for which the indicated products are defined. Determine whether the statement below is true or false. Justify the answer.

$(AB)C = (AC)B$

Choose the correct answer below.

- **A.** The statement is true. The associative law of multiplication for matrices states that $(AB)C = (AC)B$.
- **B.** The statement is false. The associative law of multiplication for matrices states that $A(BC) = (AB)C$.
- **C.** The statement is false. The associative law of multiplication for matrices states that $(AB)C = B(AC)$.
- **D.** The statement is true. The associative law of multiplication for matrices states that $(AB)C = B(AC)$.

---

10. Let A and B be arbitrary matrices for which the indicated product is defined. Determine whether the statement below is true or false. Justify the answer.

$(AB)^T = A^T B^T$

Choose the correct answer below.

- **A.** The statement is true. The transpose of the product of two matrices is the product of the transposes of the individual matrices in the same order, or $(AB)^T = A^T B^T$.
- **B.** The statement is true. Matrix multiplication is not commutative so the products must remain in the same order.
- **C.** The statement is false. The transpose of the product of two matrices is the product of the transpose of the first matrix and the second matrix, or $(AB)^T = A^T B$.
- **D.** The statement is false. The transpose of the product of two matrices is the product of the transposes of the individual matrices in reverse order, or $(AB)^T = B^T A^T$.

---

11. Determine whether the statement below is true or false. Justify the answer.

The transpose of a sum of matrices equals the sum of their transposes.

Choose the correct answer below.

- **A.** The statement is false. For matrices A and B whose sizes are appropriate to sum, $(A + B)^T = A^T + B^T$.
- **B.** The statement is false. For matrices A and B whose sizes are appropriate to sum, $(A + B)^T = A^T - B^T$.
- **C.** The statement is true. This is a generalized statement that follows from the theorem $(A + B)^T = A^T - B^T$.
- **D.** The statement is true. This is a generalized statement that follows from the theorem $(A + B)^T = A^T + B^T$.